Sheridan College, Davis Campus

SYST17796 Fundamentals of Software Design and Development

Group 10: Project Deliverable-2 Card Game War

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SYST 17796 DELIVERABLE 2
DESIGN DOCUMENT TEMPLATE

OVERVIEW

1. Project Background and Description

We implemented the classic "War" card game, in which two players receive half of a shuffled deck and compete over multiple rounds. In each round, both players play one card, and the higher card wins, awarding that player a point. After a preset number of rounds, the scores are compared, and the overall winner is declared. This game demonstrates OOP principles such as Encapsulation, Aggregation, and Composition by cleanly separating concerns like Card, Deck, Player, Round, and WarGame and implementing the UML class diagram design.

2. Design Considerations

1. **CardGameWar (Main Class):**

   This is the main class of the game. It is responsible for taking user input and entry points of the game.

   - Scanner used to take players (player 1 and 2) names as input.
   - It is used to take game total round numbers as input.
   - Creates a war game object, passing the player names and choosing the number of rounds.
   - Calling playGame() on the WarGame object to start the game.

2. **Card class:**

   We have three attributes: suit, rank, and values:

   - Suit = It stores card suit (Heart, Diamod, Clubs, Spades).
   - Rank = It stores card rank (2,3,4,…,Q,K,..).
   - Value = It represents the numeric value (2-14) that are assigned to rank.

   We have two methods and three constructor:

   - Card(String suit, String rank, int value) = Constructor.
   - getValue() = It returns the value of card.
   - toString() = It returns card details in a readable way.

3. **Deck Class:**

   Attributes:

- Cards(List<Card>): Stores all 52 cards in the deck.

Constructor:

- Deck() = Creates all 52 cards using loops. It assigns ranks, suits and values. And to randomly shuffle the deck we use collection.shuffle(cards);

Methods:

- drawCard() = It returns the top card from the deck.
- isEmpty() = It returns true if the deck is empty.

4. **Player class:**

The class represents the players in the game, like name, hand, and score

- Name, hand, and score are attributes in which the name stores the player's name, the hand has a queue storing the player's card, and the score has a total of round wins by the player.
- Methods:

addCard(), playCard(), addScore(), getScore(), getHandSize(), getName().

5. **Round class:**

This class controls a single round in the War game, holding two cards and a round number.

Attributes:

- roundNumber(): Store the sequential round index
- card1 and card2: the cards played by Players 1 and 2 in this round.

Constructor:

- Round(int roundNumber, Card card1, Card card2) – initializes the round index and the two played cards.

Method:

- compareCards(), getRoundNumber().

6. **WarGame class:**

This class controls the overall flow of the war game, which includes multiple rounds, two players, and a deck of cards.

Attributes:

- Player 1 and 2 stores player objects.

- The deck of cards
- The game will play in total
- Store each round data.

Constructor:

- WarGame(String player1Name, String player2Name, int totalRounds)
- creates two players, generates a deck object, distributes cards evenly to both players, and sets a total number of rounds.

Method:

- PlayGame() plays one round where both players draw a card, compare card values
- determineWinner() – displays the final scores and announces the overall winner.

3. UML Class Diagram